

RABINA TWAYANA

rabinatwayana123@gmail.com | [LinkedIn](#) | [Github](#) | [ePortfolio](#) | [Google Scholar](#)

Earth Observation and GeoAI researcher with 4+ years of professional experience in geospatial research, programming, and WebGIS application development. Recently completed a Copernicus Master's in Digital Earth, specializing in Geodata Science, with research focused on multimodal deep learning for wildfire spread prediction. Passionate about applying Earth Observation and geospatial intelligence to address environmental, climate, and disaster-related challenges.

EDUCATION

EMJM Copernicus Master in Digital Earth

Oct 2024 - Sept 2026

GeoData Science, Southern Brittany University (UBS), France (2025/2026)

GeoInformatics, Paris Lodron University Salzburg (PLUS), Austria (2024/2025)

Grade: 1st Year (PLUS): 1.044 out of 5 [Scale: 1=Excellent to 5=Fail]

2nd Year (UBS): 16.414 out of 20

Master's Thesis: *Advancing Wildfire Spread Prediction Through Multimodal Deep and Continual Learning*

Proposed and developed **CALM-Net** (Cross-Attention L-TAE FiLM Network), a multimodal deep learning architecture for wildfire spread prediction that explicitly models interactions between dynamic weather, temporal information, and static terrain and vegetation data. Reconstructed and improved the WSTS+ benchmark dataset and investigated domain-incremental continual learning with LoRA and experience replay to improve model adaptation to changing environmental conditions.

Selected Projects (Additional Project details: [CDE e-Portfolio](#)):

- Deep Learning-based Semantic Flood Mapping using SegFormer on Multimodal Earth Observation Data [\[Link\]](#)
- SAR-based ship detection using YOLO, DETR, and Faster R-CNN [\[Link\]](#)
- End-to-end, scalable, distributed big data machine learning pipeline [\[Link\]](#) and deep learning (UNet) pipeline [\[Link\]](#) on Apache Spark for land cover classification
- Earthquake Damage Modelling using Random Forest (RF) and Histogram Gradient Boosting (HGBC) Classifier [\[Link\]](#)
- Improved burned area detection using dual-pol Sentinel-1 data and machine learning approach: Random Forest and XGBoost [\[Link\]](#)
- Development of PySARFlow: An open-source python library for SAR data processing [\[Link\]](#)
- Development of NickySpatial: An open-source object-based image analysis library for remote sensing [\[Link\]](#)
- Hyperspectral image classification using Long Short-Term Memory (LSTM) Network [\[Link\]](#)

Bachelor's of Engineering in Geomatics

Sept 2016 - Dec 2020

Kathmandu University, Dhulikhel, Nepal

Grade: 3.86/4.00 (CGPA)

Thesis: *Land Cover Change Analysis and its Future Prediction: A Case Study of Banepa and Dhulikhel Municipalities* [\[Link\]](#)

Analyzed land-cover changes from 1992–2020 using multi-temporal Landsat remote sensing imagery, identified key drivers of land-cover change, and used the CA-ANN model in the MOLUSCE plugin in QGIS to predict the 2032 land-cover scenario.

PROFESSIONAL EXPERIENCE

Remote Sensing and AI Intern | UNU-EHS, Bonn, Germany

July-August 2024

- Developed a Retrieval-Augmented Generation (RAG) based climate knowledge retrieval web platform for AI-driven access to support climate risk and adaptation research workflows [\[Link\]](#), using RAG pipelines with OpenAI API, FastAPI, PostgreSQL, ReactJS, WebSocket, Docker, and DigitalOcean.
- Supported the CLIMADA (Climate Adaptation) Python tool upgrade by resolving complex hazard file format issues.

Research Associate | Asian Institute of Technology, Thailand

Nov 2022 - August 2024

- Led full-stack development and deployment of the [RiskChanges](#) Platform, under Deliverable 4.1 (The web-based simulation and information service for multi-hazard impact chains), using ReactJS, Leaflet, Django, REST API, PostgreSQL, Docker, and GCP, as part of the PARATUS Project, funded by the European Union's Horizon Europe research and innovation programme.

Backend/WebGIS Developer | Naxa Pvt. Ltd., Nepal**Dec 2020 - Oct 2022**

- Led backend development of multiple WebGIS applications, designing scalable APIs and databases, collaborating with frontend developers to ensure seamless integration and responsive user experience, and coordinated with international and local stakeholders to align technical implementation with project goals.
- Developed and maintained platforms using Django, REST API, PostgreSQL, GeoServer, Docker, Celery, and Redis.
- Delivered geospatial solutions for diverse stakeholders, including: Projects: (1) Clean-up Nepal [\[link\]](#), (2) WebGIS-based participatory mapping toolkit to aid evidence-based decision making ([USAFIRI](#)), (3) Provincial Climate Change Management Information System ([PCCMIS](#)), (4) Open Data Portal for Changuarayan Municipality [\[link\]](#), (5) Changuarayan Municipality Metric Addressing System, (6) Web-based application for Geodetic Division of Survey Department of Nepal, (7) Blue Infrastructure Mapping of Kathmandu Metropolitan City

Geospatial Analyst | Airlift Company, Nepal**Feb 2022 – July 2022**

- Manage the geospatial dataset for the Galli (Google) Maps Mobile Application Development of Nepal
- Processed UAV imagery using Agisoft and generated an accurate contour map to support Road Corridor Design in Bhutan

Frontend Developer Intern | Naxa Pvt. Ltd., Nepal**Jun 2020 - Aug 2020**

- Developed the frontend (user interface) for the Headquarter Mapping initiative, a nationwide open spatial data initiative to locate and map 753 Local Governments and ward Offices across Nepal

SKILLS

Programming: Python, JavaScript

Remote Sensing Data: Sentinel-1(SAR), Sentinel-2, Landsat, VIIRS, UAV/Drone Imagery

Machine Learning and Deep Learning: Scikit-learn, PyTorch, TorchGeo, Tensorflow

LLMs: Retrieval-Augmented Generation (RAG) pipelines using OpenAI API

Geospatial Software: ArcGIS, ArcGIS Online, QGIS, ENVI, Agisoft, Geoserver

Geospatial and Remote Sensing: WebGIS, GIS, UAV, GNSS, GEE

Data analysis & visualization: Matplotlib, Seaborn

Distributed Computing / Big Data: PySpark

Frontend: ReactJs, Redux, Openlayers, Leaflet, Cesium,

Backend: Django, REST API, FastAPI

Database Management: PostgreSQL

Containerization and Version Control: Docker, Git, Github

Cloud Infrastructure: Google Cloud Platform (GCP), Amazon Web Services (AWS), DigitalOcean

Operating System: Linux, Windows, Mac

Language: English, Nepali (Mother Tongue)

PUBLICATION

Han, L., Huang, W., Zhang, X., Shi, Y., Dong, Y., Sobeih, T., Metta, C., **Twayana, R.**, Parkhedkar, G., Wang, D., Patel, K. A., Sok, K., Tran Khanh, D., Mohanta, S., & Shashwat, S. (2026). ICPR 2026 Competition on Beyond Visible Spectrum: AI for Agriculture. *Pattern Recognition, Lecture Notes in Computer Science*, 16827, 276–290. https://doi.org/10.1007/978-3-032-31936-4_19

Twayana, R., & Hadj-Rabah, K. (2026). Assessment of Dual-Polarization Sentinel-1 SAR Data for Improved Wildfire Burned Area Mapping: A Case Study of the Palisades Region, USA. *Geomatics*, 6(2), 28. <https://doi.org/10.3390/geomatics6020028>

Bhandari, S., **Twayana, R.**, Shrestha, R., & Sharma, K. (2021). Future Land Use Land Cover Scenario Simulation Using Open-Source GIS for the City of Banepa and Dhulikhel Municipality, Nepal. *Proceedings of FOSS4G-Asia 2021*, 73–83. https://www.geoinfo-lab.org/foss4g-asia_2021/wp-content/uploads/2022/09/FOSS4G-Asia-2021-Proceedings_Final.pdf

Twayana, R., Bhandari, S., & Shrestha, R. (2021). Analyzing Urban Growth Pattern and Driving Factors Using Remote Sensing and GIS: A Case Study of Banepa Municipality, Nepal. *Journal on Geoinformatics, Nepal*, 20(1), 9–18. <https://doi.org/10.3126/njg.v20i1.39471>

Shrestha, R., Bhandari, S., & **Twayana, R.** (2021). Monitoring Land Cover/Land Use Change using Remote Sensing and GIS Technique at Dhulikhel and Banepa Municipality, Nepal. *International Research Journal of Modernization in Engineering Technology & Science*, 03(05). https://www.irjmets.com/uploadedfiles/paper/volume3/issue_5_may_2021/9793/1628083392.pdf

CONFERENCE PRESENTATION

Twayana, R., Akintola, O. M., Ogallo, E., & Peres, B. (2025, October 27). PySARFlow: An Open-Source Python Library for SAR Data Processing. In *2025 ASPRS International Technical Symposium* [Online]. <https://community.asprs.org/2025symposium/symposium-program/technical-sessions/2802>

Twayana, R., Westen, C. V., Hazarika, M. K., & Shakya, A. R. (2023, December 7). RiskChanges: A WebGIS Application for Multi-Hazard Quantitative Risk Assessment. In *3rd International Symposium on Disaster Resilience and Sustainable Development (DRSD 2023)*, Asian Institute of Technology, Thailand [Onsite][[link](#)].

OUTREACH

Twayana, R. (7 November 2023). An open-source tool to assist in multi-hazard risk assessment. *DRR Community Voices*. <https://www.preventionweb.net/drr-community-voices/open-source-tool-assist-multi-hazard-risk-assessment>

HONOURS AND AWARDS

Best Student Award, Copernicus Masters in Digital Earth, 2026

Based on academic excellence, Copernicus relevance, student engagement and outreach during the academic years 2024/2025 - 2025/2026

2nd Place, ICPR 2026 Competition on Beyond Visible Spectrum AI for Agriculture (Task 2), 2026

Implemented MoCo (Momentum Contrast) for self-supervised learning (SSL) and fine-tuned ResNet for downstream crop disease classification [[Github](#)] [[Paper](#)]

Erasmus Mundus Scholarship, 2024

Awarded full scholarship for joint master's program, Copernicus Master in Digital Earth at Paris Lodron University Salzburg, Austria and specialization in GeoData Science at Southern Brittany University, France

Certificate of Honor, Nepal Institute of Chartered Surveyors (NICS), 2021

For securing the second position in Bachelor of Geomatics Engineering degree examination conducted by all Universities of Nepal

Kathmandu University, 2016-2020

Waved 100% tuition fee in four semesters during undergraduate study for scoring the highest GPA in the respective semester

Land Management Training Centre (LMTC), Government of Nepal, 2016

Waved 35 % of tuition fee by LMTC for four years undergraduate course (BE in Geomatics) at Kathmandu University, Nepal.

Private and Boarding School's Organization Nepal, Kavrepalanchowk District Executive Committee, 2014

For securing a distinction division with 86.38% marks in the SLC Exam

TRAINING AND PARTICIPATION

AI4EO 20026 Spring School, University of Southern Brittany, 8-10 April 2026, Vannes, France

Machine learning in Python with scikit-learn, massive open online course (MOOC), completed on 28 October 2025

AI4EO Symposium, 11-12 September 2025, Rennes, France

Using Drones in Environmental Sciences, Blended Intensive Programme (BIP), February-June 2025, onsite (2-6 June 2025), University of Bucharest, Romania

Make an Impact with Modern Geo Apps (MOOC), 4 weeks, online, completed on May 2 2025

Going Places with Spatial Services (MOOC), 6 weeks, online, completed on March 6 2025

Introduction to Remote Sensing and Hands-on Disaster Management Applications, October 28 - November 27 2024, self-paced and virtual training by SpaceSUITE

Business Strategy and Business Model, self-paced online course by Space Academy, completed on 17 Nov 2024

Planet on the Road: Bangkok, 26 September 2023, Bangkok, Thailand

VISESS Summer School in Risk and Resilience of 6 ECTS held on 10-14 July 2023 at the University of Vienna, Austria

ASEAN Training on ICT for Climate Resilient Development, organized by APCICT, ESCAP, 15-17 May 2023, Bangkok, Thailand

VOLUNTEERING EXPERIENCE

Volunteer (24 & 25 June), Living Planet Symposium, 2025, Vienna, Austria

Facilitator in Flood Risk Assessment Training for Bhutan's National Center for Hydrology and Meteorology, held from 21st Aug to 1st Sept 2023 at Geoinformatics Center, Asian Institute of Technology, Thailand

Editorial Board Member, Annual Geo-ITC Journal of Geomatics Engineering Society (ISSN 2091-198), Kathmandu University, Nepal